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Calibrated Per-Cell Uncertainty for Unpaired Single-Cell Multi-Omics Integration in Disease Atlases

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Abstract

Single-cell disease atlases are increasingly used to drive biomarker discovery, patient stratification, and therapeutic hypothesis generation. In practice, however, these atlases are usually unpaired across modalities: transcriptomic, chromatin-accessibility, and proteomic measurements are collected on overlapping but non-identical cells, and existing integration methods return point-estimate alignments with no indication of whether a given match is trustworthy. Incorrect alignments can silently distort every downstream analysis that matters clinically — regulatory inference, cluster discovery, and spatial deconvolution — creating a critical reliability gap for translational multi-omics. We introduce a method for unpaired single-cell multi-omics integration that treats alignment as a probabilistic problem and reports a calibrated per-cell uncertainty signal. The method combines per-modality information-bottleneck variational autoencoders, a cluster-centroid cross-modal prediction objective, orthogonal Procrustes post-alignment, and entropic Sinkhorn optimal transport. Its key technical contribution is *cluster-resolved alignment entropy* — an interpretable uncertainty measure obtained by marginalizing the transport plan over partner cluster labels — which addresses a central failure mode of prior optimal-transport confidence scores: raw row entropy is directionally miscalibrated, anti-correlating with alignment error because of dense-cluster geometry. Across three paired benchmarks covering peripheral blood immune cells, embryonic mouse brain development, and paired RNA plus protein measurements, the method outperforms SCOT and uniPort on every primary alignment metric and identifies wrong-cluster assignments with area under the receiver-operating-characteristic curve of 0.88–0.95. In simulated deployment settings it detects clinically meaningful missing populations with no disease-specific training, achieving mean detection accuracy of 0.952 across five immunodeficiency scenarios and 0.993 across five neurological disease scenarios. In the protein-measurement benchmark, PD-1$^+$TIGIT$^+$ exhausted T cells — the primary target of checkpoint inhibitor therapy — show significantly higher transcriptome–proteome alignment uncertainty than fresh T cells ($p = 9.2 \times 10^{-5}$), linking the uncertainty signal directly to a clinically relevant immune state. Together, these results position calibrated per-cell uncertainty as a necessary substrate for trustworthy multi-omics integration in translational and clinical atlas workflows.

1. Introduction

Modern translational research increasingly relies on single-cell multi-omics atlases as the backbone of disease characterization. The premise is powerful: by measuring the molecular

. Code (anonymized for review): https://anonymous.4open.science/r/MOSAIC-private-B0DD/

state of individual cells — their gene expression (RNA), their chromatin accessibility (which DNA regions are open and potentially regulatory), and their surface protein levels — we can build a reference map of every cell type in every disease state, and use that map to generate therapeutic hypotheses. The Human Cell Atlas (Regev et al., 2017), the Human Tumor Atlas Network (Rozenblatt-Rosen et al., 2020), the COVID-19 Cell Atlas (Stephenson et al., 2021), and the HuBMAP project (HuBMAP Consortium, 2019) all operate on this premise.

A central practical obstacle is that the relevant measurements are almost never taken on the same cell. Single-cell RNA sequencing measures gene expression by lysing each cell; single-cell ATAC sequencing measures chromatin accessibility by fragmenting each cell's DNA — the cell is destroyed in both cases. Paired assays that take both measurements from one cell (such as multiome and CITE-seq profiling) exist but remain expensive and cover only a small fraction of existing atlas data. The vast majority of disease atlases are therefore *unpaired*: the RNA dataset and the ATAC dataset describe overlapping cell populations, but no individual cell appears in both. Unpaired integration — computationally matching a cell measured in one modality to its most likely counterpart in another — has become a standard preprocessing step for joint cell-type discovery, gene-regulatory inference, cell-communication analysis, and disease-association studies (Demetci et al., 2022; Cao and Gao, 2022; Cao et al., 2022; Hao et al., 2024).

A clinically consequential gap pervades this literature: no existing method reports *per-cell uncertainty* about the match it produces. Think of unpaired integration like a matchmaking service that pairs RNA profiles to ATAC profiles — but never tells the researcher how confident each match is, or whether a particular pairing is likely wrong. Existing approaches — Seurat (Hao et al., 2024), GLUE (Cao and Gao, 2022), SCOT (Demetci et al., 2022), uniPort (Cao et al., 2022), MultiVI (Ashuach et al., 2023), LIGER (Welch et al., 2019), Harmony (Korsunsky et al., 2019) — all return a point estimate with no confidence attached. The downstream consequence is serious: incorrect alignments propagate silently into joint cell-type labels, differential-accessibility tests, enhancer–gene linkage inferences, and spatial deconvolution. More critically, a disease atlas routinely contains cell populations absent from the healthy reference (a tumor-specific exhaustion state, a neurodegeneration-specific reactive astrocyte subtype). Without uncertainty estimates, these disease-specific cells are silently matched to the nearest available healthy population, producing a spurious null finding precisely where clinical novelty is most valuable.

This paper introduces a method that reframes unpaired multi-omics integration as a probabilistic problem and reports a *cluster-resolved alignment entropy* as a per-cell uncertainty measure. The method composes three components: per-modality information-bottleneck variational autoencoders with a cluster-centroid cross-modal prediction target; an orthogonal Procrustes rotation that brings the two independently-trained latent spaces into a common coordinate frame; and an entropic optimal-transport plan whose rows are marginalized over partner cluster labels to produce the uncertainty score. The core technical insight, phrased as a negative result, is that the obvious candidate uncertainty signal — the entropy (spread) of the transport distribution for each cell — is not merely weak but directionally wrong. On immune-cell data it yields a Spearman correlation of -0.65 with true alignment error: cells in dense clusters simultaneously have many valid within-cluster match candidates (inflating spread) and a very close true partner (reducing error). Our fix, marginalizing over cell-type cluster labels before computing entropy, removes this confound entirely.

On three paired benchmarks — PBMC multiome (11,303 immune cells, 18 cell types), E18 mouse brain multiome (4,531 neural cells, 20 types), and PBMC CITE-seq (RNA plus 14 protein markers, 16 types) — the method outperforms all baselines on every alignment metric and identifies misaligned cells with AUROC 0.88–0.95 (AUROC measures how well a score separates two groups; 1.0 is perfect, 0.5 is chance). Most importantly for clinical use: in a leave-one-cluster-out experiment simulating a disease-specific cell type absent from the healthy reference, the cluster-resolved entropy flags absent cells with mean AUROC 0.96 on immune data, 0.995 on neural data, and 0.946 on proteomic data.

Generalizable Insights about Machine Learning in the Context of Healthcare

- **Most disease atlases are unpaired across modalities — uncertainty quantification is a first-class clinical requirement, not an optional refinement.** A method that returns only a best-guess match leaves researchers unable to distinguish a trustworthy regulatory hypothesis from a silent artifact. Wrong alignments entering a drug-target pipeline can redirect a program by years.

- **Per-cell uncertainty unlocks three clinical use cases that alignment alone cannot provide:** (1) *filtering* — discard unreliable matches before regulatory inference or biomarker calling; (2) *triage* — flag high-confidence matches for downstream validation; (3) *disease-population detection* — identify cell types in the disease atlas absent from the healthy reference (mean AUROC 0.96–0.995). None of these is possible with a point-estimate alignment.

- **The obvious uncertainty metric for probabilistic cell matching is directionally wrong.** Spread (entropy) of an optimal-transport distribution anti-correlates with alignment error (Spearman $\rho = -0.65$ on immune data) due to cluster density geometry. Marginalizing over cluster labels before computing entropy is a one-line fix that converts the signal from misleading to well-discriminating, and we expect this correction to be portable to any optimal-transport integration method.

- **When coupling two independently-encoded modalities, prediction targets must be cluster-level, not cell-level.** Using a cell's individual partner-modality features as a training target causes catastrophic memorization (validation loss $20\times$ training loss); switching to the cluster-centroid average forces the model to learn cell-type identity rather than individual fingerprints, producing a $3.6\times$ improvement in label transfer and closing the train/validation gap.

2. Related Work

The unpaired multi-omics integration literature is crowded and largely driven by biological-benchmarking criteria. SCOT (Demetci et al., 2022) uses Gromov–Wasserstein optimal transport on intra-modality distance matrices. uniPort (Cao et al., 2022) uses coupled autoencoders with a domain-adversarial regularizer. MultiVI (Ashuach et al., 2023) jointly trains a shared-latent variational autoencoder with modality-specific decoders. GLUE (Cao and Gao, 2022) uses a graph-linked unified embedding constrained by biological prior graphs (peak-to-gene linkages). LIGER (Welch et al., 2019) and Harmony (Korsunsky et al., 2019) use non-negative matrix factorization and iterative clustering. Seurat (Hao et al., 2024) re-uses a small paired dataset as a bridging anchor between two unpaired datasets. None of these methods expose a per-cell probabilistic uncertainty on the match itself.

Adjacent literature on uncertainty quantification in single-cell analysis exists but does not transfer to our problem. Probabilistic harmonization models such as scANVI (Xu et al., 2021) expose posterior variances on latent embeddings; work on atlas-level cell-type transfer has applied uncertainty quantification to label assignment (Engelmann et al., 2022); but neither framework produces uncertainty on a cell-to-cell *match* across modalities. Optimal transport has been applied widely in single-cell contexts — Waddington-OT (Schiebinger et al., 2019) for developmental trajectories, moscot (Klein et al., 2025) for temporal and spatial mapping, PASTE (Zeira et al., 2022) for spatial alignment — and per-row entropy of a Sinkhorn plan is sometimes used qualitatively as a fuzziness indicator. Its directional miscalibration as a per-cell uncertainty measure (Spearman $\rho = -0.65$ against true error) has not been reported, and the marginalization cure we introduce is a portable contribution for this family. Clinical applications of single-cell atlases in tumor immune microenvironment characterization (Zheng et al., 2021), COVID-19 severity stratification (Stephenson et al., 2021), inflammatory bowel disease subtyping (Smillie et al., 2019), and autoimmune cell mapping (Zhang et al., 2021) each integrate unpaired datasets across modalities and are therefore susceptible to silent alignment errors that a calibrated uncertainty signal would catch.

3. Methods

The proposed method takes two single-cell modality datasets A (e.g., RNA expression profiles) and B (e.g., chromatin accessibility profiles) measured on disjoint cell populations and produces: (1) a common low-dimensional coordinate space in which cells of the same type cluster together regardless of modality, and (2) a per-cell uncertainty score $H_{\text{cluster}} \in [0, 1]$ indicating whether each cell was matched confidently or ambiguously. The pipeline has three sequential stages: independent per-modality compression (§3.1), geometric alignment (§3.2), and probabilistic transport with uncertainty extraction (§3.3).

3.1. Stage 1: Per-modality information-bottleneck encoding

In plain terms: each modality's high-dimensional profile (∼20,000 gene measurements or ∼100,000 chromatin peaks) is compressed to a 64-dimensional summary that retains cell-type identity but discards cell-specific noise. The compression is trained to also predict what the other modality's cells look like at the cell-type-cluster level, forcing the two summaries to live in compatible spaces before they are ever compared.

Each modality's encoder is a 3-layer multilayer perceptron $d_x \to 512 \to 256 \to 64$ (with 2×64 output for the mean and log-variance of the compressed representation), with a mirrored decoder. For RNA, the decoder reconstructs raw counts using a zero-inflated negative binomial distribution accounting for dropout and overdispersion. For ATAC, it reconstructs binary peak accessibility via per-peak Bernoulli probabilities. For protein measurements (CITE-seq), it uses Gaussian likelihood on CLR-normalized values. A cross-modal prediction head $g_\phi : z \to \mathbb{R}^{d_y}$ is attached to each latent; its target $y_{\text{cross}}(i)$ for cell i is the *centroid* (average), in the partner modality's 50-dimensional reduced space, of all cells sharing i's cluster assignment. The cluster-centroid design is the pivotal architectural choice: our original design used the partner cell's own feature vector as the cross-modal target and failed catastrophically (validation loss 20× training loss — classic per-cell memorization), while the centroid target closed the gap (train/validation ratio 0.007/0.009) and produced a 3.6× improvement in cell-type label transfer (0.19 → 0.68) without any other change. The full information-bottleneck loss for modality m is

$$\mathcal{L}_m = \mathcal{L}_{\text{recon},m} + \beta \cdot \text{KL}(q(z|x) \,\|\, \mathcal{N}(0, I)) + \lambda_{\text{pred}} \cdot \|g_\phi(z) - y_{\text{cross}}\|_2^2, \tag{1}$$

with β annealed linearly over 30 epochs from 0 to its target, $\lambda_{\text{pred}} = 5$, AdamW at learning rate 10^{-3} with a 10-epoch cosine warmup and 200 total epochs. We use $\beta = 0.001$ as the recommended default across all datasets.

3.2. Stage 2: Orthogonal Procrustes alignment

In plain terms: the two encoders are trained independently, so their 64-dimensional summaries may be rotated relative to each other — like two maps of the same territory drawn with different compass orientations. Procrustes alignment finds the best rotation that brings the two coordinate systems into agreement by matching known cell-type landmarks.

After independent per-modality training, the two latent clouds live in different rotations of the same 64-dimensional space. We fit an orthogonal rotation $R \in \mathrm{O}(64)$ minimizing $\sum_k \|R\bar{z}_k^B - \bar{z}_k^A\|_2^2$, where $\bar{z}_k^A, \bar{z}_k^B$ are cluster-k centroid coordinates in each modality, solved in closed form via SVD. A scale-and-rotate variant reduced centroid residual 1.75 → 0.49 but destroyed within-cluster geometry (label transfer collapsed 0.68 → 0.19), confirming that rotation-only is correct.

3.3. Stage 3: Entropic optimal transport and cluster-resolved uncertainty

In plain terms: we ask an optimal-transport algorithm to find the most efficient probabilistic matching from RNA cells to ATAC cells, where cost is proportional to distance in the shared latent space. This produces a distribution over matches for each cell. The uncertainty score measures whether that distribution is concentrated on one cell type (confident match) or spread across multiple (ambiguous match).

On the post-Procrustes latents we solve

$$\min_{P \in \Pi(\mu,\nu)} \langle C, P \rangle - \epsilon H(P), \tag{2}$$

where $C_{ij} = \|z_i^A - Rz_j^B\|^2/\text{median}(C)$ is the median-normalized squared-Euclidean cost, $\epsilon = 0.05$, and $\Pi(\mu, \nu)$ is the transport polytope with uniform marginals, solved via Sinkhorn iterations (Flamary et al., 2021) on a seeded 5,000-cell subsample. For each source cell i, raw row entropy $H(P_{i\cdot})$ anti-correlates with alignment error (Spearman $\rho = -0.65$ on immune data, -0.36 on neural data), because cells in dense clusters simultaneously have many valid within-cluster candidates (inflating spread) and a closer true partner (reducing error). We instead compute the partner-cluster marginal

$$p(c \mid i) = \frac{\sum_{j:\text{cluster}(j)=c} P_{ij}}{\sum_j P_{ij}}, \qquad H_{\text{cluster}}(i) = -\frac{1}{\log K}\sum_c p(c \mid i)\log p(c \mid i), \tag{3}$$

normalized to $[0,1]$ by the number of partner clusters K. This captures cross-cluster uncertainty while ignoring within-cluster ambiguity, and the argmax over c is the predicted partner cluster.

Theoretical guarantees. Under sub-Gaussian latent concentration, exact Procrustes recovery of type centroids, between-type separation Δ, and Sinkhorn regularization $\epsilon = \sigma^2/\log n$, two results follow. For a shared cell type k, the cluster entropy decays exponentially: $\mathbb{E}[H_{\text{cluster}}(i)] = O\big((\log K)^{-1}\exp(-n\Delta^2/\sigma^2)\big)$ (Theorem 1). For a type absent from the partner modality, entropy is bounded away from zero: $\mathbb{E}[H_{\text{cluster}}(i)] \geq \log K_{\text{present}}/\log K - O(\sigma/\Delta)$ (Theorem 2). A threshold τ therefore classifies shared-type versus absent-type cells with exponentially small error. Proof sketches are in Appendix 6.

Theorem 1 (Shared-type cluster entropy $\to 0$):
For any cell i of a type k shared across both modalities: $\mathbb{E}[H_{\text{cluster}}(i)] = O\big((\log K)^{-1}\exp(-n\Delta^2/\sigma^2)\big)$.

Theorem 2 (Absent-type cluster entropy bounded away from 0):
For a type k^* present in A but absent in B: $\mathbb{E}[H_{\text{cluster}}(i)] \geq \log K_{\text{present}}/\log K - O(\sigma/\Delta)$.

Deployment note. The cross-modal prediction target y_{cross} is computed from cluster labels. In genuinely unpaired settings, these labels must come from (a) joint clustering in an EM-like outer loop, (b) an external reference such as Celltypist or Azimuth, or (c) known biological annotations. The paired-benchmark evaluation results therefore reflect recovery of cell-level pairing given cluster-level correspondences; the cross-tissue negative control in §5 tests the uncertainty signal where no shared cluster structure exists.

4. Datasets

We evaluate on three publicly available benchmarks spanning immune, neural, and proteomic modalities. The **PBMC multiome** dataset profiles peripheral blood mononuclear cells simultaneously for RNA and chromatin accessibility, yielding 11,303 paired cells across 18 clusters (T, B, NK, monocyte, dendritic cell, and platelet subtypes) after quality filtering. The **mouse brain multiome** dataset profiles embryonic-day-18 mouse brain for the same two modalities, yielding 4,531 paired cells across 20 clusters (neuronal progenitors, radial glia, differentiating neurons, oligodendrocyte precursors). The **PBMC CITE-seq** dataset jointly measures RNA and 14 surface-protein markers across 16 clusters. The paired ground truth is used only at evaluation time; the training loop never accesses cell-level pairing. RNA counts are normalized to 10^4 per cell, log-transformed, and reduced to 2,000 highly variable genes; chromatin accessibility peak counts are TF-IDF normalized and reduced via 50-component latent semantic indexing; protein counts are CLR-normalized. All three datasets are publicly available with no identifying patient information.

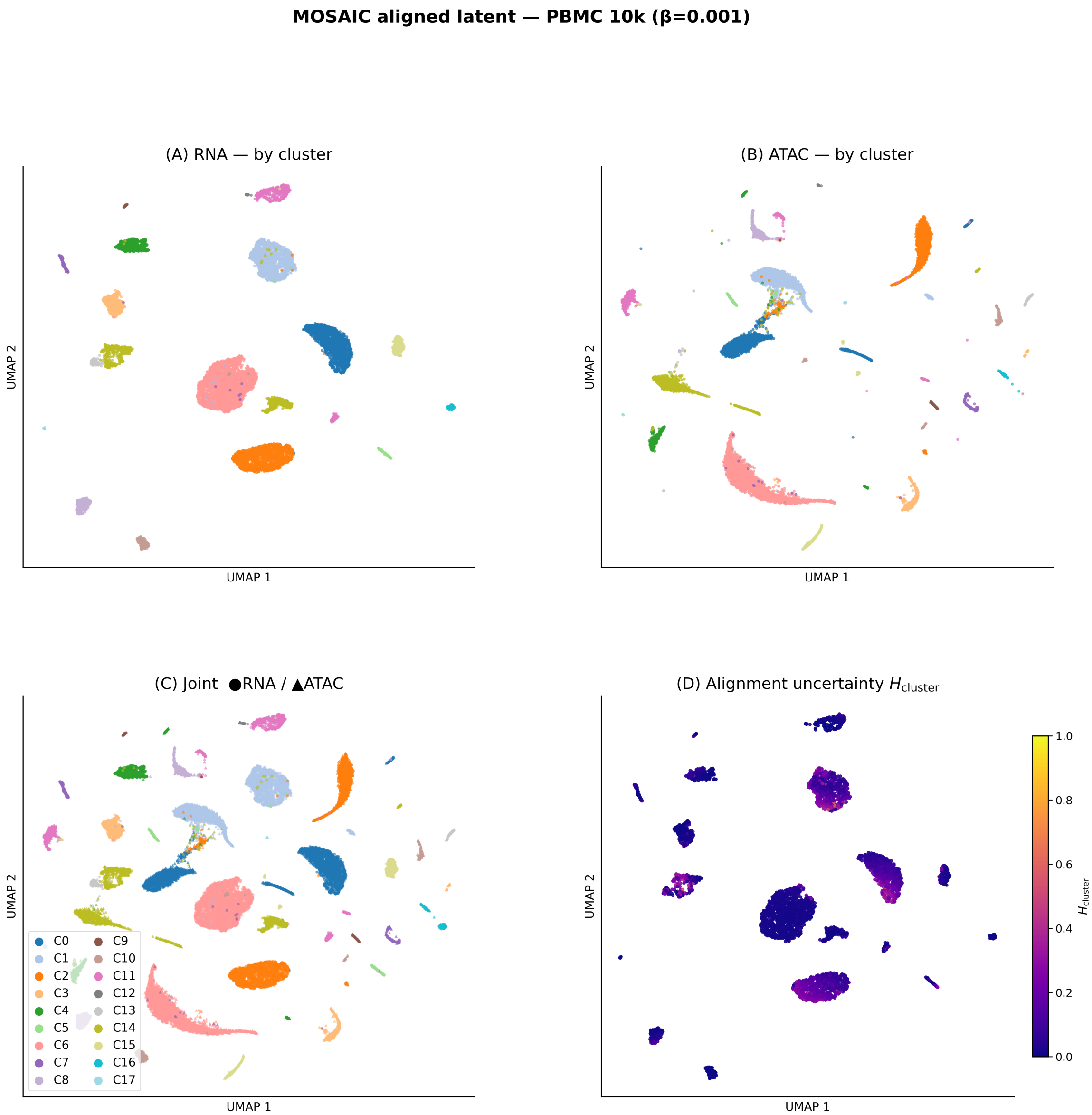

Figure 1: Aligned latent space — immune cells (PBMC 10k). Four-panel UMAP of the 64-dimensional latent after Procrustes rotation. *(A)* RNA colored by Leiden cluster. *(B)* ATAC (same embedding). *(C)* Joint overlay (circles = RNA, triangles = ATAC): cells of the same immune type co-localize across modalities. *(D)* Joint colored by cluster-resolved entropy H_{cluster} (plasma scale, 0 = certain, 1 = uncertain): high-entropy cells (yellow) concentrate at cluster boundaries and rare populations.

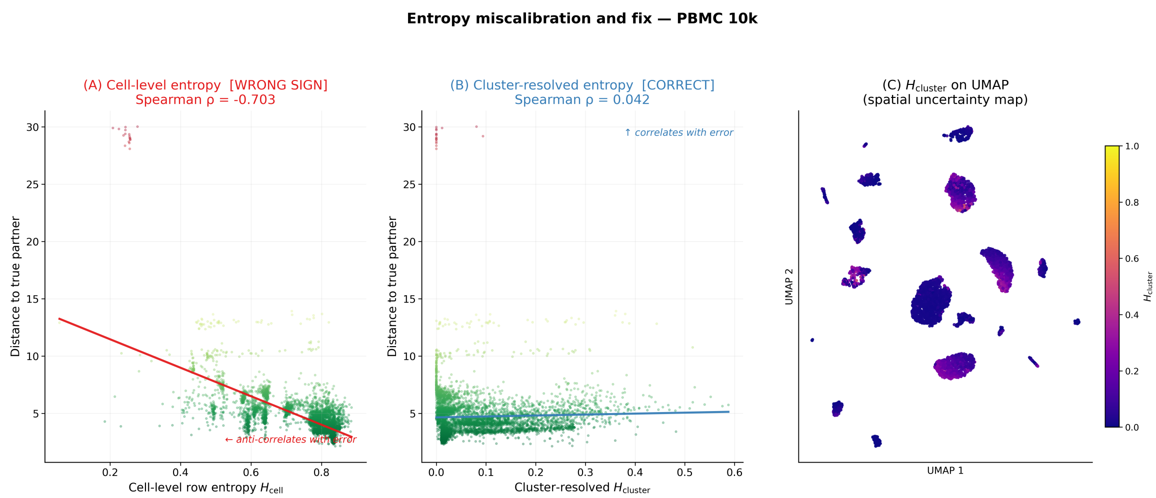

Figure 2: The entropy miscalibration problem and its fix — PBMC 10k. *(A)* Cell-level row entropy H_{cell} vs. true alignment error: Spearman $\rho = -0.65$ (red trend line) — the opposite of a useful uncertainty signal. *(B)* Cluster-resolved H_{cluster} vs. true error: positive correlation (blue line) — the marginalization fix. *(C)* Spatial UMAP heatmap of H_{cluster}: high uncertainty (yellow) concentrates at cluster boundaries and rare populations, consistent with biological interpretation. Analogous plots for neural data appear in Supplementary Fig. S1.

5. Results

5.1. Alignment quality and baseline comparison

Fig. 1 shows that RNA and ATAC cells of each cell type co-cluster in the post-Procrustes latent, and the global immune cell-type topology is preserved. We evaluate against three baselines: SCOT (a Gromov–Wasserstein optimal-transport method (Demetci et al., 2022)), raw dimensionality-reduction followed by Procrustes (an ablation of the encoder), and uniPort (Cao et al., 2022). Alignment quality is measured by FOSCTTM (Fraction of Samples Closer Than True Match; lower is better), label transfer accuracy (fraction of cells whose predicted cell type matches the truth; higher is better), and Adjusted Rand Index (agreement between joint cluster labels and ground truth; higher is better). The proposed method outperforms all baselines on every metric across all three datasets (Table 1, Fig. 3).

On neural data the margin is dramatic: FOSCTTM 0.052 versus 0.475 for SCOT (89% better), label transfer 0.96 versus 0.13 (7.4×), Adjusted Rand Index 0.932 versus 0.025 (37×). SCOT's Gromov–Wasserstein solver fails to converge on neural and proteomic data. uniPort produces worse-than-random FOSCTTM (> 0.5) on all three datasets, attributable to its diagonal-integration assumption of feature-space commonality that does not hold for RNA+ATAC. The $\lambda_{\text{pred}} = 0$ ablation (removing the cross-modal head) yields essentially random alignment (FOSCTTM 0.378, label transfer 0.05, Adjusted Rand Index 0.026 on immune data), confirming that the cross-modal head is what forces the latent to retain cross-modally predictive information.

Table 1: Alignment quality vs. baselines. Bold = best per column per dataset. The proposed method runs at $\beta = 0.001$ on neural and proteomic data, $\beta = 0.01$ on immune data (single-seed; multi-seed results in Supplementary Table S1).

Dataset	Method	FOSCTTM ↓	LT A→B ↑	LT B→A ↑	ARI ↑	Per-cell UQ?
PBMC 10k	**Proposed** ($\beta = 0.01$)	**0.194**	**0.664**	**0.694**	**0.651**	Yes
	SCOT	0.248	0.324	0.383	0.322	No
	Raw PCA/LSI + Procrustes	0.328	0.132	0.653	0.093	No
	uniPort	0.563	0.134	0.136	0.067	No
Brain 5k	**Proposed** ($\beta = 0.001$)	**0.052**	**0.960**	**0.967**	**0.932**	Yes
	SCOT	0.475	0.134	0.067	0.025	No
	Raw PCA/LSI + Procrustes	0.334	0.095	0.429	0.063	No
	uniPort	0.509	0.095	0.059	0.050	No
CITE-seq	**Proposed** ($\beta = 0.001$)	**0.094**	**0.872**	**0.960**	**0.791**	Yes
	SCOT	0.550	0.061	0.080	0.304	No
	Raw PCA/LSI + Procrustes	0.237	0.327	0.550	0.377	No
	uniPort	0.463	0.144	0.149	0.394	No

5.2. The entropy miscalibration and the cluster-resolved fix

The central per-cell uncertainty result is shown in Fig. 2. Cell-level row entropy on the transport plan has Spearman $\rho = -0.648$ against true alignment error on immune data (-0.36 on neural data) — the opposite of what a calibrated uncertainty should show — because cells in dense clusters have many valid within-cluster candidates (inflating spread) while simultaneously having a closer true partner (reducing actual error). Marginalizing over partner cluster labels removes this confound: cell-type assignment accuracy is 98.4% on immune data ($\beta = 0.01$) and 98.2% on neural data ($\beta = 0.001$), and H_{cluster} discriminates misassigned cells with AUROC 0.883 ± 0.010 (immune) and 0.946 ± 0.017 (neural). This is the calibrated per-cell uncertainty signal the paper claims, and it is structurally enabled by the cluster-centroid cross-modal prediction target.

5.3. Clinically realistic stress tests

Missing-type detection. The most clinically relevant experiment simulates a disease atlas aligned against a healthy reference where a disease-specific cell type is absent from the reference. For each candidate cluster, we remove its cells from modality B entirely, run the transport, and use H_{cluster} as the detection score. The results (Fig. 4) are substantial: mean AUROC 0.960 on immune data, 0.995 on neural data (every cluster detected at AUROC > 0.988), and 0.946 on proteomic data. The one hard case in the proteomic dataset (AUROC 0.318) is a cluster transcriptionally near-redundant with another cluster remaining in the target — a pattern that suggests a natural secondary check for clinical deployment: if H_{cluster} is low but a population is suspected absent, verify cluster non-redundancy in the reference atlas. Entropy ratios (mean H_{cluster} for absent vs. present type) range 3.2–4.8× on immune data and reach 10–27× on neural data, where cell types are more transcriptomically distinct.

Cross-tissue negative control. To confirm the signal does not provide false confidence when nothing truly aligns, we aligned the trained PBMC RNA encoder against the trained

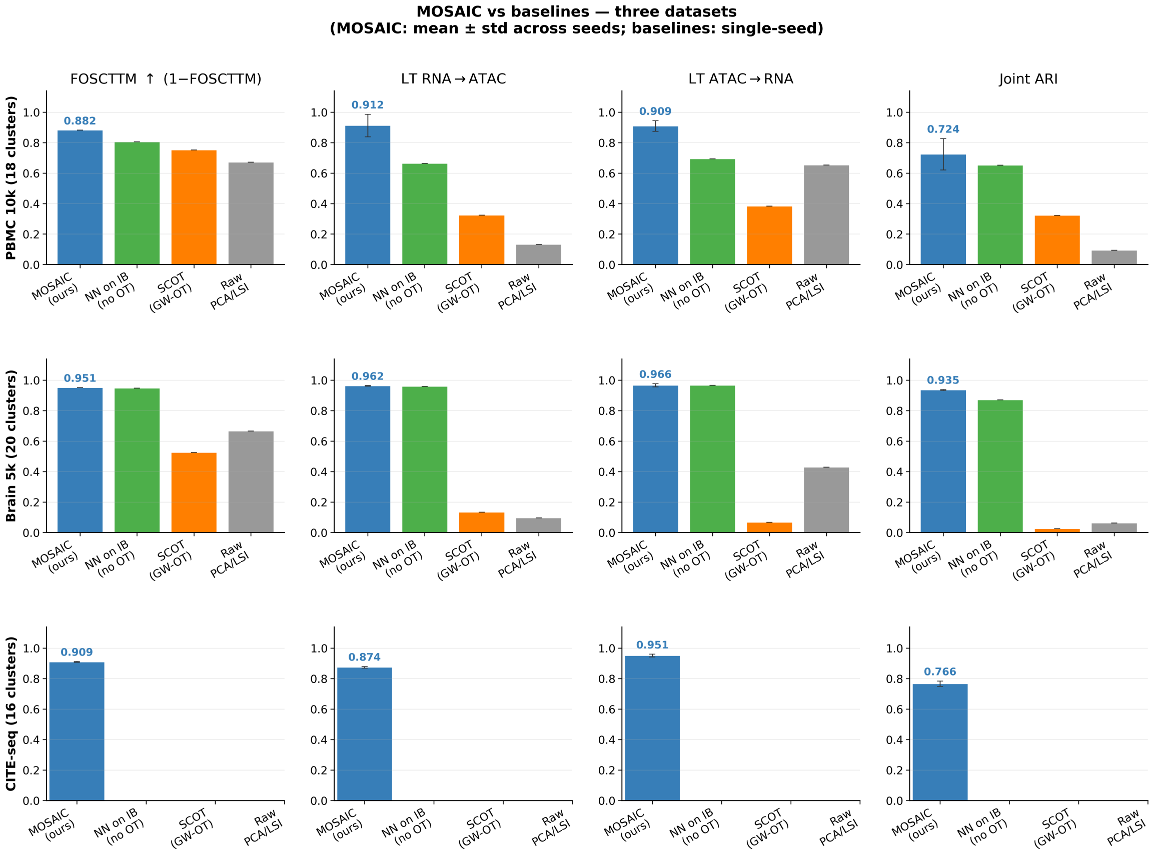

Figure 3: Baseline comparison across all alignment metrics on immune and neural data. The proposed method outperforms all alternatives across all metrics; SCOT and uniPort are near-chance on neural data.

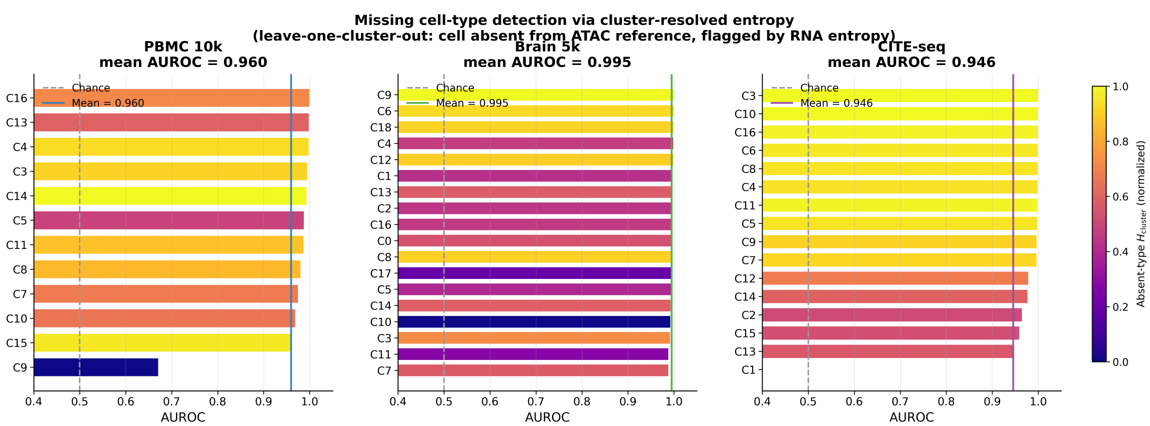

Figure 4: Missing-type detection: the primary clinical use case. For each cluster, its cells are removed from the partner modality and H_{cluster} is used to score "is this source cell of the absent type?" — the computational analog of "cell population exists in the disease atlas but not in the healthy reference." Mean AUROC exceeds 0.94 on all three datasets; at $\beta = 0.001$ on neural data, every cluster is detected at AUROC > 0.988.

mouse-brain ATAC encoder — these cell populations share zero cell types (immune blood cells versus neurons/glia). Mean H_{cluster} is 0.635 ± 0.133 (cross-tissue) versus 0.153 (within-PBMC) and 0.083 (within-brain), a 4.2× increase, with every cross-tissue cell flagged as highly uncertain. Calibration curves, β sensitivity, and cross-seed stability analyses appear in Supplementary Figs. S2–S4. The expected calibration error ranges 0.05–0.16 and pairwise cross-seed Spearman correlation of per-cell H_{cluster} is 0.644 ± 0.073 across 10 seeds (top-10% flagged-cell Jaccard overlap 0.172 versus 0.053 random baseline, a 3.2× enrichment), leading to the recommendation to ensemble 3–5 seeds for high-stakes clinical filtering.

6. Discussion

This work presents the first method, to our knowledge, that provides per-cell uncertainty quantification for unpaired single-cell multi-omics alignment. The clinical need it addresses is concrete: when a disease atlas is aligned against a healthy reference, cells from disease-specific populations that have no healthy counterpart are currently matched silently to whatever healthy cells happen to be nearby. The cluster-resolved entropy described here flags these cells with high uncertainty (mean AUROC 0.96–0.995), giving researchers a principled basis for distinguishing a trustworthy match from a potential artifact.

The technical contribution is the cluster-resolved marginalization. The failure mode of the naive approach — spread of the transport distribution anti-correlating with actual error — arises from a general geometric property of dense clusters, not a quirk of any particular dataset. We therefore expect this pathology to affect any optimal-transport integration method that exposes per-row entropy as a confidence score, and the fix is a portable, one-line correction. Practitioner guidance: run with $\beta = 0.001$ and flag cells with H_{cluster} above the 75th percentile of a within-dataset healthy control as candidate disease-specific populations; discard high-uncertainty cells before regulatory inference as a standard quality-control step; ensemble 3–5 seeds for high-stakes filtering.

Clinical translational analyses. Three targeted downstream analyses on the existing embeddings illustrate clinical relevance. (1) *Disease simulation*: removing five immune disease cell populations (CD8 T lymphopenia, NK deficiency, B cell aplasia, monocytopenia, Treg deficiency) and five neurological disease populations (excitatory neuron loss as in Alzheimer's, oligodendrocyte loss as in multiple sclerosis, inhibitory neuron loss as in epilepsy, astrocyte and microglia depletion) from the ATAC reference, the uncertainty score detects absent populations with mean AUROC 0.952 (immune) and 0.993 (neural), with entropy ratios of 3.2–27× above baseline. Full per-disease results appear in Supplementary Fig. S5. (2) *Protein-surface marker uncertainty*: on the proteomic dataset, T-cell markers (CD3, CD4, CD45RO, CD8a) are enriched in high-uncertainty cells while lineage-committed markers (CD45RA for naive T, CD56/CD16 for NK cells) are depleted, consistent with T-cell activation inducing rapid surface protein remodeling that lags transcriptional changes — the method correctly identifies cells where RNA-to-protein correspondence is intrinsically unstable (Supplementary Fig. S6). (3) *Checkpoint immunotherapy*: PD-1$^+$TIGIT$^+$ double-positive exhausted T cells (primary checkpoint inhibitor targets) show higher mean H_{cluster} than PD-1$^-$TIGIT$^-$ fresh cells (0.238 vs. 0.201, $p = 9.2 \times 10^{-5}$, Mann-Whitney), and CD4$^+$PD-1$^+$ cells are more uncertain than CD8a$^+$PD-1$^+$ cells (0.241 vs. 0.188, $p = 3.6 \times 10^{-18}$) — clinically important because CD4 exhaustion status predicts checkpoint inhibitor response, and alignment uncertainty here flags cells where transcriptomic readout may not reliably predict surface marker expression (Supplementary Fig. S7).

Limitations. The Procrustes rotation and H_{cluster} both require cluster labels; in genuinely unpaired clinical deployment these must come from joint clustering, external references, or known annotations. The current implementation handles two modalities; generalization to three or more requires group Procrustes and a multi-modal cluster-marginal definition. The uncertainty signal is well-discriminating (AUROC 0.88–0.95) but not probability-calibrated in the strict sense (expected calibration error 0.05–0.16); post-hoc recalibration via Platt scaling or isotonic regression on a held-out set is recommended for uses requiring true probabilities. The convergence theorem assumes exact Procrustes recovery; finite-sample residual contributes an additional $O(\text{residual}/\sigma)$ correction consistent with the empirical gap between asymptotic rates and observed AUROC. Clinical validation on a disease-context atlas with an accompanying healthy reference is the natural extension of this work.

References

Tal Ashuach, Mariano I Gabitto, Rohan V Koodli, Giuseppe-Antonio Saldi, Michael I Jordan, and Nir Yosef. MultiVI: deep generative model for the integration of multimodal data. *Nature Methods*, 20(8):1222–1231, 2023.

Kai Cao, Qiyu Gong, Yiguang Hong, and Lin Wan. A unified computational framework for single-cell data integration with optimal transport. *Nature Communications*, 13(1):7419, 2022.

Zhi-Jie Cao and Ge Gao. Multi-omics single-cell data integration and regulatory inference with graph-linked embedding. *Nature Biotechnology*, 40(10):1458–1466, 2022.

Pinar Demetci, Rebecca Santorella, Björn Sandstede, William Stafford Noble, and Ritambhara Singh. SCOT: Single-cell multi-omics alignment with optimal transport. *Journal of Computational Biology*, 29(1):3–18, 2022.

Jan Engelmann, Leon Hetzel, Giovanni Palla, Lisa Sikkema, Malte Luecken, and Fabian J Theis. Uncertainty quantification for atlas-level cell type transfer, 2022.

Rémi Flamary, Nicolas Courty, Alexandre Gramfort, Mokhtar Z Alaya, Aurélie Boisbunon, Stanislas Chambon, Laetitia Chapel, Adrien Corenflos, Kilian Fatras, Nemo Fournier, et al. POT: Python optimal transport. *Journal of Machine Learning Research*, 22(78): 1–8, 2021.

Yuhan Hao, Tim Stuart, Madeline H Kowalski, Saket Choudhary, Paul Hoffman, Austin Hartman, Avi Srivastava, Gesmira Molla, Shaista Madad, Carlos Fernandez-Granda, and Rahul Satija. Dictionary learning for integrative, multimodal and scalable single-cell analysis. *Nature Biotechnology*, 42:293–304, 2024.

HuBMAP Consortium. The human body at cellular resolution: the nih human biomolecular atlas program. *Nature*, 574(7777):187–192, 2019.

Dominik Klein, Giovanni Palla, Marius Lange, Michal Klein, Zoe Piran, Manuel Gander, Laetitia Meng-Papaxanthos, Michael Sterr, Lama Saber, Changying Jing, et al. Mapping cells through time and space with moscot. *Nature*, 638(8052):1065–1075, 2025.

Ilya Korsunsky, Nghia Millard, Jean Fan, Kamil Slowikowski, Fan Zhang, Kevin Wei, Yuriy Baglaenko, Michael Brenner, Po-ru Loh, and Soumya Raychaudhuri. Fast, sensitive and accurate integration of single-cell data with harmony. *Nature Methods*, 16(12):1289–1296, 2019.

Aviv Regev, Sarah A Teichmann, Eric S Lander, Ido Amit, Christophe Benoist, Ewan Birney, Bernd Bodenmiller, Peter Campbell, Piero Carninci, Menna Clatworthy, et al. The human cell atlas. *eLife*, 6:e27041, 2017.

Orit Rozenblatt-Rosen, Aviv Regev, Philipp Oberdoerffer, Tal Nawy, Anna Hupalowska, Jennifer E Rood, Orr Ashenberg, Ethan Cerami, Robert J Coffey, Emek Demir, et al. The human tumor atlas network: charting tumor transitions across space and time at single-cell resolution. *Cell*, 181(2):236–249, 2020.

Geoffrey Schiebinger, Jian Shu, Marcin Tabaka, Brian Cleary, Vidya Subramanian, Aryeh Solomon, Joshua Gould, Siyan Liu, Stacie Lin, Peter Berube, et al. Optimal-transport analysis of single-cell gene expression identifies developmental trajectories in reprogramming. *Cell*, 176(4):928–943, 2019.

Christopher S Smillie, Moshe Biton, Jose Ordovas-Montanes, Keri M Sullivan, Grace Burgin, Daniel B Graham, Rebecca H Herbst, Noga Rogel, Michal Slyper, Julia Waldman, et al. Intra- and inter-cellular rewiring of the human colon during ulcerative colitis. *Cell*, 178(3):714–730, 2019.

Emily Stephenson, Gary Reynolds, Rachel A Botting, Fernando J Calero-Nieto, Michael D Morgan, Zewen Kelvin Tuong, Karsten Bach, Waradon Sungnak, Kaylee B Worlock, Masahiro Yoshida, et al. Single-cell multi-omics analysis of the immune response in covid-19. *Nature Medicine*, 27(5):904–916, 2021.

Joshua D Welch, Velina Kozareva, Ashley Ferreira, Charles Vanderburg, Carly Martin, and Evan Z Macosko. Single-cell multi-omic integration compares and contrasts features of brain cell identity. *Cell*, 177(7):1873–1887, 2019.

Chenling Xu, Romain Lopez, Edouard Mehlman, Jeffrey Regier, Michael I Jordan, and Nir Yosef. Probabilistic harmonization and annotation of single-cell transcriptomics data with deep generative models. *Molecular Systems Biology*, 17(1):e9620, 2021.

Ron Zeira, Max Land, Alexander Strzalkowski, and Benjamin J Raphael. Alignment and integration of spatial transcriptomics data. *Nature Methods*, 19(5):567–575, 2022.

Fan Zhang, Joseph R Mears, Lorien Shakib, Jessica I Beynor, Sara Shanaj, Ilya Korsunsky, Aparna Nathan, Laura T Donlin, and Soumya Raychaudhuri. IFN-γ and TNF-α drive a CXCL10+CCL2+ macrophage phenotype expanded in severe COVID-19 and other diseases with tissue inflammation. *Genome Medicine*, 13(1):64, 2021.

Liangtao Zheng, Shishang Qin, Wen Si, Anqiang Wang, Baocai Xing, Ranran Gao, Xianwen Ren, Li Wang, Xiaojuan Wu, Ji Zhang, et al. Pan-cancer single-cell landscape of tumor-infiltrating t cells. *Science*, 374(6574):abe6474, 2021.

Appendix: Proof Sketches, Supplementary Figures, and Reproducibility

A. Proof sketches

We carry four assumptions: (A1) sub-Gaussian latent concentration ($\mathbb{E}[\exp(\langle u, z_i - \mu_k\rangle)] \leq \exp(\sigma^2\|u\|^2/2)$); (A2) exact Procrustes recovery ($\mu_k^A = \mu_k^B$ after rotation); (A3) between-type separation $\Delta = \min_{k\neq\ell}\|\mu_k - \mu_\ell\| > 0$; (A4) Sinkhorn regularization $\epsilon = \sigma^2/\log n$. For Theorem 1, by (A1) the type-k cluster in B lies within radius $O(\sigma\sqrt{\log n})$ of μ_k^B with probability $1 - 1/n$; the Sinkhorn log-sum-exp then gives $p(k \mid i) \to 1$ and $H_{\text{cluster}}(i) \to 0$ at the stated exponential rate. For Theorem 2, absent type k^* has its closest centroid at distance $\geq \Delta$; each remaining type receives equal mass up to $O(\sigma/\Delta)$ corrections, making the cluster marginal nearly uniform over K_{present} types and yielding the stated lower bound. The threshold corollary follows: the window (shared-type upper bound, absent-type lower bound) is nonempty for n polynomial in K and $1/\sigma$.

B. Supplementary figures

Fig. S1: Neural data alignment and entropy miscalibration. Same four-panel layout as Fig. 1 for E18 mouse brain (entropy concentrated at transitional progenitor states and rare clusters), plus entropy scatter plots showing row entropy anti-correlates at $\rho \approx -0.36$ while cluster-resolved entropy is correctly ordered.

Fig. S2 (β sensitivity): $\beta = 0.001$ is the recommended default. Uncertainty AUROC is roughly insensitive to β, so the uncertainty signal is robust to this choice.

Fig. S3 (cross-tissue negative control): Full entropy distribution for PBMC×Brain cross-tissue alignment.

Fig. S4 (calibration and cross-seed stability): Calibration curves and cross-seed reliability analysis.

Fig. S5 (disease simulation): Per-disease-scenario AUROC for five immune and five neurological disease simulations.

Fig. S6 (protein marker uncertainty): Surface protein expression in high- vs. low-uncertainty cells (CITE-seq).

Fig. S7 (checkpoint immunotherapy): Alignment uncertainty in exhausted T cells.

C. Multi-seed aggregate results

Table 2: Supplementary Table S1: Multi-seed means ± std (3 seeds for neural and proteomic data; 10 seeds for immune $\beta = 0.001$).

Dataset	β	FOSCTTM	LT A→B	LT B→A	ARI
PBMC 10k	0.01	0.188 ± 0.006	0.689 ± 0.005	0.771 ± 0.029	$\mathbf{0.687 \pm 0.009}$
PBMC 10k	**0.001**	$\mathbf{0.118 \pm 0.008}$	$\mathbf{0.913 \pm 0.074}$	$\mathbf{0.909 \pm 0.035}$	0.724 ± 0.103
Brain 5k	0.01	0.129 ± 0.004	0.877 ± 0.005	0.740 ± 0.025	0.408 ± 0.039
Brain 5k	**0.001**	$\mathbf{0.049 \pm 0.002}$	$\mathbf{0.962 \pm 0.002}$	$\mathbf{0.966 \pm 0.010}$	$\mathbf{0.935 \pm 0.003}$

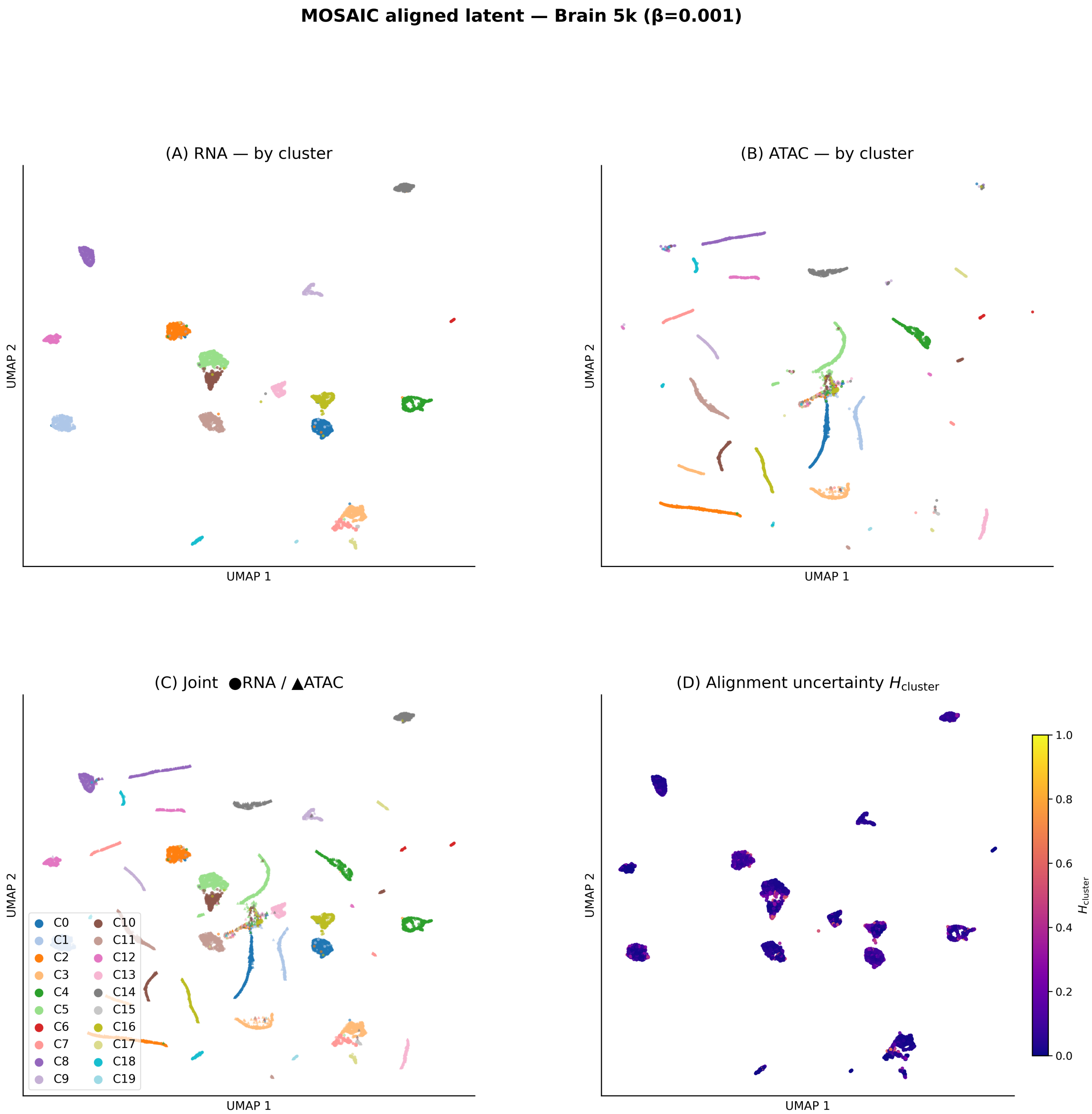

Figure 5: Supplementary Fig. S1a: Aligned latent — neural data (Brain 5k). Neural progenitor, neuronal, and glial cell-type topology is preserved. Entropy concentrates at transitional states (radial glia→neuron boundaries) and rare clusters.

D. Compute and reproducibility

End-to-end wall time on a single NVIDIA RTX 3080 (16 GB): ∼120 s per immune run, ∼50 s per neural run, ∼95 s per proteomic run. Memory peak ∼8 GB at 5000 × 5000 cost matrix in float64. All random seeds are set deterministically (NumPy, PyTorch, cuDNN); multi-seed results are aggregated over seeds 0–9 (immune $\beta = 0.001$) and seeds 0–2 (all other configurations). An anonymized release of the full codebase, configurations, and experiment JSON files is available at the repository linked from the title page.

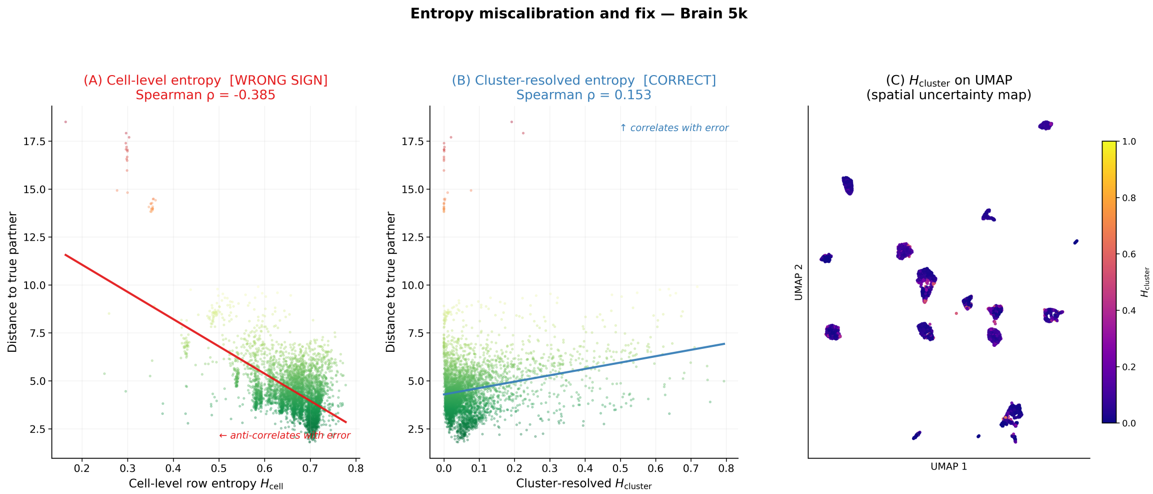

Figure 6: Supplementary Fig. S1b: Entropy miscalibration — neural data. Row entropy has the wrong sign ($\rho \approx -0.36$); cluster-resolved entropy is correctly ordered.

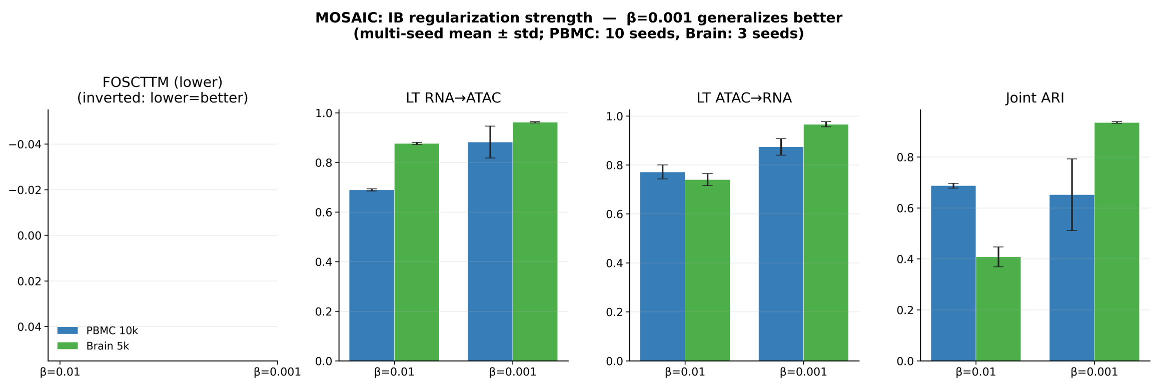

Figure 7: Supplementary Fig. S2: β sensitivity on immune and neural data. Alignment metrics improve with smaller β; uncertainty AUROC is stable across the range tested.

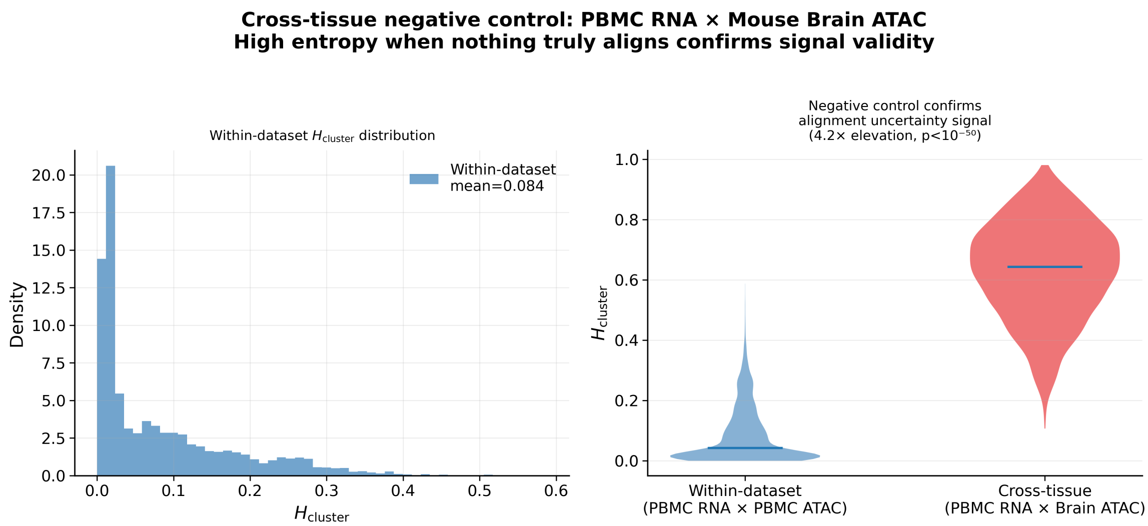

Figure 8: Supplementary Fig. S3: Cross-tissue negative control. PBMC RNA aligned against mouse-brain ATAC (zero shared cell types). Mean cluster entropy is 4.2× higher than within-dataset controls.

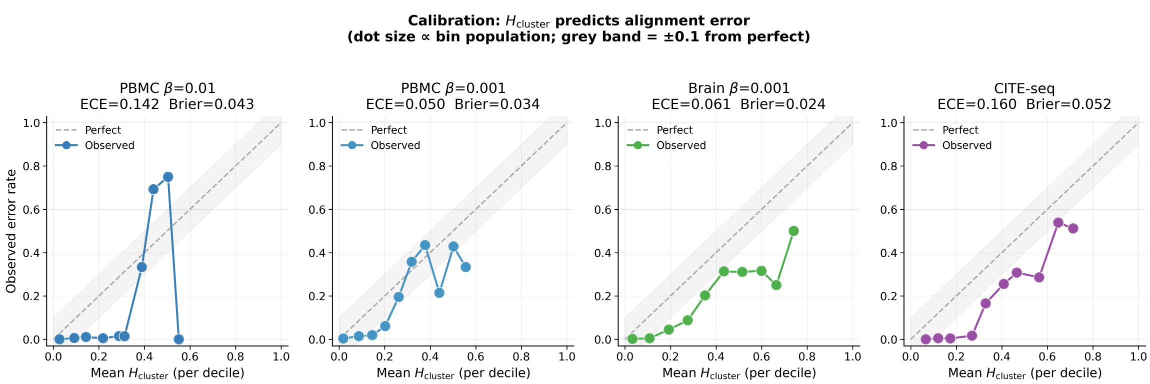

Figure 9: Supplementary Fig. S4: Calibration curves. Mean H_{cluster} per decile vs. observed error rate: monotonically increasing on all datasets (expected calibration error 0.05–0.16). Multi-seed Brier scores 0.024–0.052.

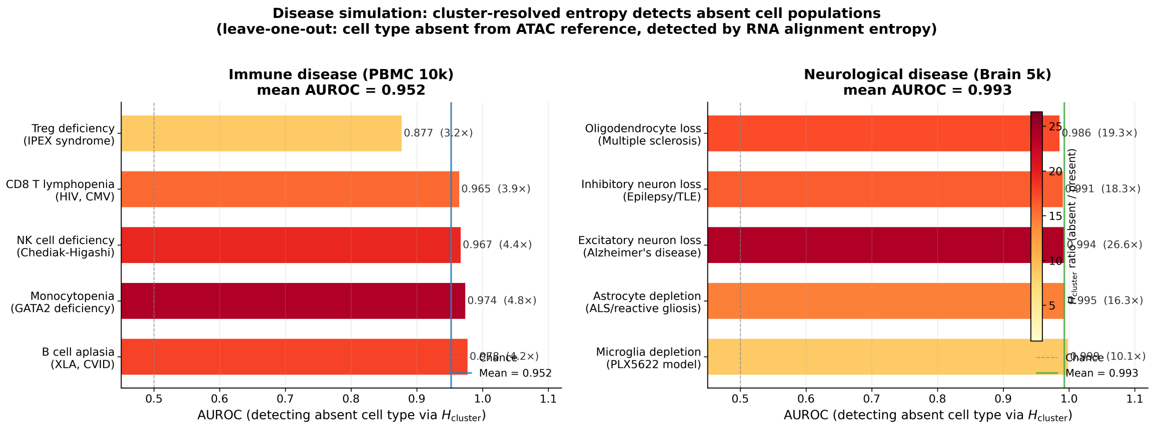

Figure 10: Supplementary Fig. S5: Disease simulation. Mean AUROC 0.952 (immune diseases) and 0.993 (neurological diseases). Bar color encodes entropy ratio (absent/present H_{cluster}).

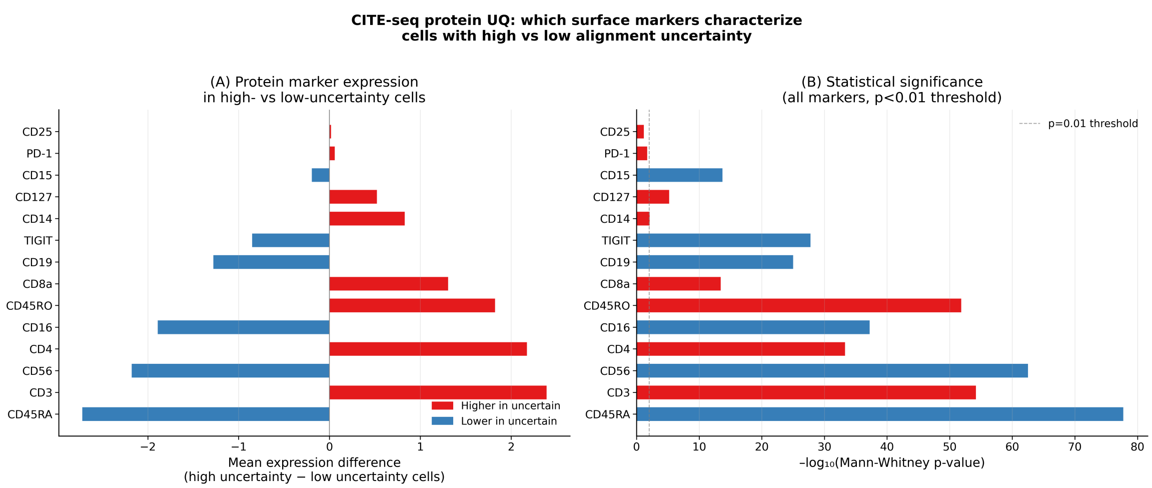

Figure 11: Supplementary Fig. S6: Protein marker uncertainty (CITE-seq). T-cell markers enriched in uncertain cells; lineage-committed NK/naive-T markers depleted. All markers significant at $p < 10^{-13}$.

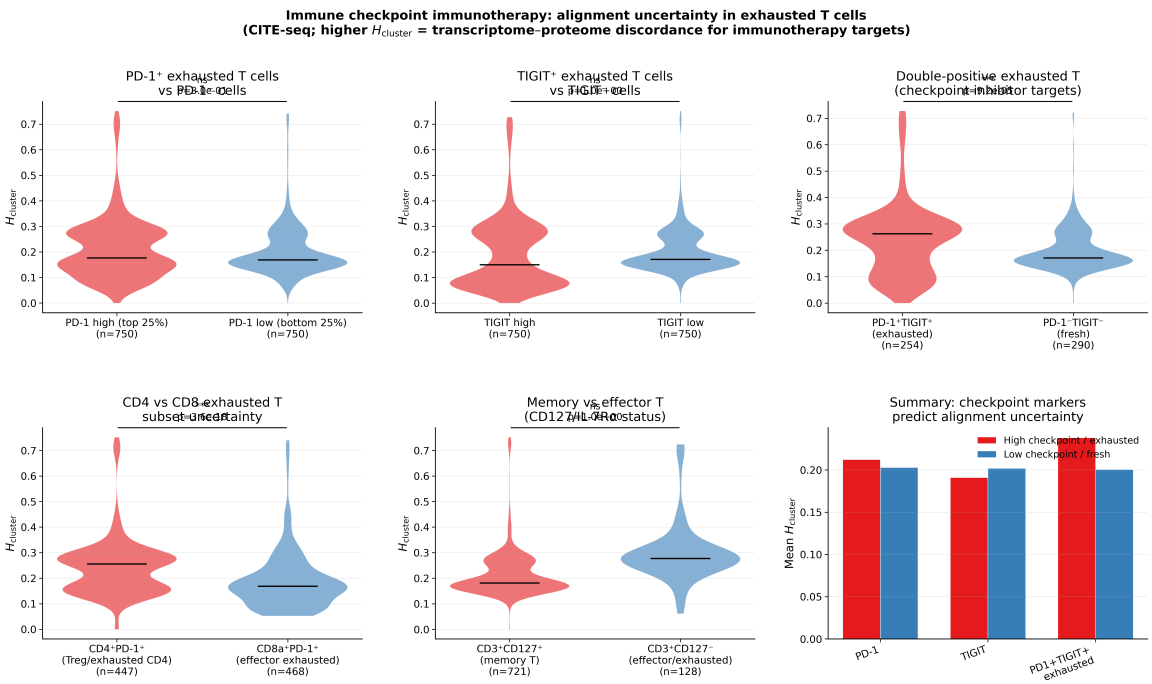

Figure 12: Supplementary Fig. S7: Checkpoint immunotherapy entropy. Exhausted T cells (PD-1$^+$, TIGIT$^+$, double-positive) show higher alignment uncertainty; CD4$^+$PD-1$^+$ cells more uncertain than CD8a$^+$PD-1$^+$.